Cost-Volume-Profit (CVP) Analysis

Comprehensive 50 MCQ Practice Bank (Theory & Numericals)

1. Which of the following is the fundamental formula for calculating the Break-Even Point (in units)?

A. Fixed Costs ÷ Total Sales
B. Fixed Costs ÷ Contribution Margin per Unit
C. Variable Costs ÷ Contribution Margin per Unit
D. Total Costs ÷ Profit

Answer: B

2. Contribution margin is defined as:

A. Sales revenue minus fixed costs
B. Sales revenue minus total costs
C. Sales revenue minus variable costs
D. Fixed costs minus variable costs

Answer: C

3. If selling price is Rs. 50, variable cost is Rs. 30, and fixed costs are Rs. 100,000, what is the break-even point in units?

A. 2,000 units
B. 3,333 units
C. 5,000 units
D. 10,000 units

Answer: C *(100,000 ÷ 20 = 5,000)*

4. The Profit/Volume (P/V) ratio indicates the relationship between:

A. Contribution and Sales
B. Profit and Fixed Costs
C. Sales and Variable Costs
D. Fixed Costs and Contribution

Answer: A

5. If the P/V ratio is 40% and sales are Rs. 500,000, what is the total contribution?

A. Rs. 100,000
B. Rs. 200,000
C. Rs. 300,000
D. Rs. 500,000

Answer: B *(500,000 × 0.40 = 200,000)*

6. Margin of Safety is best described as:

A. The difference between actual sales and break-even sales
B. The difference between total sales and total costs

7. A company has actual sales of Rs. 800,000 and its break-even sales are Rs. 500,000. What is the Margin of Safety?

A. Rs. 300,000
B. Rs. 500,000
C. Rs. 1,300,000
D. 37.5%

Answer: A *(800,000 - 500,000)*

8. What happens to the break-even point if fixed costs increase, assuming all other factors remain constant?

A. It decreases
B. It remains the same
C. It increases
D. It becomes negative

Answer: C

9. If a firm's Margin of Safety is 20% and its actual sales are 50,000 units, what are the break-even sales in units?

A. 10,000 units
B. 40,000 units
C. 50,000 units
D. 60,000 units

Answer: B *(50,000 - 20% = 40,000)*

10. At the break-even point, which of the following is true?

A. Total Revenue = Variable Costs
B. Total Contribution = Fixed Costs
C. Total Profit = Margin of Safety
D. Fixed Costs = Variable Costs

Answer: B

11. A firm desires a target profit of Rs. 50,000. Fixed costs are Rs. 150,000 and unit contribution margin is Rs. 20. How many units must be sold?

A. 7,500 units
B. 10,000 units
C. 12,500 units
D. 15,000 units

Answer: B *((150,000 + 50,000) ÷ 20 = 10,000)*

12. Which of the following is NOT an assumption of CVP analysis?

A. Costs can be accurately classified into fixed and variable
B. Selling price per unit remains constant
C. Sales mix changes continuously
D. Volume is the only factor affecting costs

14. Using the data from Question 13 (P/V Ratio = 30%, Year 1 Sales = 100,000, Profit = 10,000), what are the total fixed costs?

A. Rs. 20,000
B. Rs. 30,000
C. Rs. 40,000
D. Rs. 50,000

Answer: A *(Contribution = 100,000 × 0.30 = 30,000; Fixed Costs = 30,000 - 10,000 = 20,000)*

15. The angle of incidence in a break-even chart represents:

A. The rate of fixed cost absorption
B. The rate at which profits are earned after break-even
C. The proportion of variable costs to fixed costs
D. The ratio of debt to equity

Answer: B

16. If a product's selling price is Rs. 200, P/V ratio is 25%, what is its variable cost per unit?

A. Rs. 50
B. Rs. 100
C. Rs. 150
D. Rs. 175

Answer: C *(Contribution = 50; Variable Cost = 200 - 50 = 150)*

17. Operating leverage measures:

A. How sensitive profit is to a change in sales volume
B. How debt affects net income
C. The ratio of fixed assets to current assets
D. The time taken to break even

Answer: A

18. Total contribution is Rs. 300,000 and net operating income is Rs. 100,000. What is the Degree of Operating Leverage (DOL)?

A. 0.33
B. 1.5
C. 2.0
D. 3.0

Answer: D *(DOL = Contribution ÷ Net Operating Income = 300,000 ÷ 100,000)*

19. If Degree of Operating Leverage is 4, a 10% increase in sales will result in what percentage increase in net income?

A. 10%
B. 20%
C. 40%
D. 400%

21. A decrease in the selling price per unit will:

A. Increase the P/V ratio
B. Decrease the break-even point
C. Increase the break-even point
D. Have no effect on the margin of safety

Answer: C

22. If fixed costs are Rs. 120,000, variable costs are 60% of sales, what is the break-even revenue?

A. Rs. 200,000
B. Rs. 300,000
C. Rs. 400,000
D. Rs. 600,000

Answer: B *(P/V Ratio = 40%. BEP = 120,000 ÷ 0.40 = 300,000)*

23. In a multi-product firm, the break-even point is calculated using:

A. Highest contribution margin
B. Lowest contribution margin
C. Weighted average contribution margin
D. Simple average contribution margin

Answer: C

24. What is the margin of safety if profit is Rs. 40,000 and the P/V ratio is 20%?

A. Rs. 8,000
B. Rs. 50,000
C. Rs. 160,000
D. Rs. 200,000

Answer: D *(Profit ÷ P/V Ratio = 40,000 ÷ 0.20 = 200,000)*

25. Which cost changes proportionately with the level of activity?

A. Fixed Cost
B. Variable Cost
C. Step Cost
D. Sunk Cost

Answer: B

26. If the Margin of Safety is high, it indicates that:

A. The company is at a high risk of incurring losses
B. A small drop in sales will lead to massive losses
C. The company has a strong cushion against falling sales
D. Fixed costs are extremely high

Answer: C

27. Selling Price: Rs. 100. Variable Cost: Rs. 40. Fixed Cost: Rs. 180,000. Tax Rate: 40%. Calculate

28. Which of the following formulas correctly calculates the Profit/Volume (P/V) ratio?

A. (Sales - Fixed Costs) ÷ Sales
B. (Contribution ÷ Sales) × 100
C. (Fixed Costs ÷ Contribution) × 100
D. (Profit ÷ Variable Costs) × 100

Answer: B

29. An increase in fixed costs will cause the Margin of Safety to (assuming sales remain constant):

A. Increase
B. Decrease
C. Remain unchanged
D. Cannot be determined

Answer: B

30. A company sells 10,000 units at Rs. 50 each. Variable costs are Rs. 30 per unit, and fixed costs are Rs. 150,000. What is the net profit?

A. Rs. 50,000
B. Rs. 100,000
C. Rs. 150,000
D. Rs. 200,000

Answer: A *(Contribution = 10,000 × 20 = 200,000. Profit = 200,000 - 150,000 = 50,000)*

31. What is the indifference point in CVP analysis?

A. The point where total sales equal total costs
B. The sales volume where profits of two alternatives are equal
C. The point where margin of safety is zero
D. The volume where fixed costs equal variable costs

Answer: B

32. Option A has Fixed Cost Rs. 50,000 & Variable Cost Rs. 10. Option B has Fixed Cost Rs. 80,000 & Variable Cost Rs. 5. What is the indifference point?

A. 3,000 units
B. 4,000 units
C. 5,000 units
D. 6,000 units

Answer: D *((80,000 - 50,000) ÷ (10 - 5) = 30,000 ÷ 5 = 6,000)*

33. If variable cost as a percentage of sales increases, the break-even point will:

A. Decrease
B. Increase
C. Stay the same
D. Become zero

Answer: B

35. Cost-Volume-Profit analysis is a technique primarily useful for:

A. Long-term strategic planning
B. Short-term profit planning and decision making
C. Valuing inventory on the balance sheet
D. Determining depreciation methods

Answer: B

36. Actual Sales Rs. 1,000,000. Margin of Safety is 40%. P/V ratio is 25%. What is the Profit?

A. Rs. 100,000
B. Rs. 150,000
C. Rs. 200,000
D. Rs. 400,000

Answer: A *(MOS = 400,000. Profit = MOS × P/V = 400,000 × 25% = 100,000)*

37. If fixed costs decrease by 10% and unit selling price increases by 10%, what will happen to the break-even point?

A. It will definitely increase
B. It will definitely decrease
C. It will remain unchanged
D. It depends on variable costs

Answer: B

38. Contribution margin can also be calculated as:

A. Fixed Costs + Net Profit
B. Fixed Costs - Net Profit
C. Variable Costs + Net Profit
D. Sales - Fixed Costs

Answer: A

39. A company sells Product A (CM = Rs. 10) and Product B (CM = Rs. 15) in a 2:3 ratio. What is the weighted average contribution margin?

A. Rs. 12.00
B. Rs. 12.50
C. Rs. 13.00
D. Rs. 14.50

Answer: C *((10 × 2/5) + (15 × 3/5) = 4 + 9 = 13)*

40. In a graphical representation of CVP analysis, the break-even point is where:

A. The fixed cost line intersects the variable cost line
B. The total revenue line intersects the total cost line
C. The profit line intersects the fixed cost line
D. The contribution margin line intersects the vertical axis

Answer: B

42. A factory produces 5,000 units. Fixed costs are Rs. 100,000, variable costs are Rs. 150,000, and profit is Rs. 50,000. What is the selling price per unit?

A. Rs. 40
B. Rs. 50
C. Rs. 60
D. Rs. 70

Answer: C *(Total Revenue = 100,000 + 150,000 + 50,000 = 300,000. SP = 300,000 ÷ 5,000 = 60)*

43. Which of the following describes "Target Costing" in relation to CVP?

A. Setting costs based on a desired profit and a competitive selling price
B. Determining profit after actual costs are calculated
C. Increasing selling price to cover all fixed costs
D. Ignoring variable costs to hit a target

Answer: A

44. A company desires to reduce its break-even point. It should:

A. Increase fixed costs
B. Decrease selling price
C. Increase variable cost per unit
D. Increase selling price

Answer: D

45. If the P/V ratio is 20% and fixed costs are Rs. 40,000, what sales volume is required to earn a profit of Rs. 20,000?

A. Rs. 100,000
B. Rs. 200,000
C. Rs. 300,000
D. Rs. 400,000

Answer: C *((40,000 + 20,000) ÷ 0.20 = 300,000)*

46. The cash break-even point is calculated by:

A. Subtracting non-cash expenses (like depreciation) from total fixed costs, then dividing by unit contribution
B. Adding non-cash expenses to total variable costs
C. Ignoring all fixed costs
D. Subtracting current liabilities from fixed costs

Answer: A

47. A company with high operating leverage:

A. Has high variable costs and low fixed costs
B. Has high fixed costs and low variable costs
C. Is less risky during an economic downturn
D. Cannot calculate a break-even point

Answer: B

49. Sales = Rs. 200,000. Profit = Rs. 20,000. Variable Costs = Rs. 140,000. What is the Margin of Safety?

A. Rs. 33,333

B. Rs. 66,667

C. Rs. 100,000

D. Rs. 133,333

Answer: B *(Contribution = 60,000. P/V = 30%. MOS = Profit ÷ P/V = 20,000 ÷ 30% = 66,666.67)*

50. An underlying limitation of CVP analysis is that it assumes:

A. Inventories change drastically from period to period

B. Technology and production efficiency remain constant

C. Costs cannot be separated into fixed and variable

D. Prices change with the volume of production

Answer: B

End of Practice Questions

Use this workbook to test your CVP and Break-Even Analysis knowledge. Review the calculation hints provided next to the answers for numerical problems.